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"NEXUS": AI-POWERED PLATFORM FOR BRIDGING THE SKILL GAP IN RURAL AND SEMI-URBAN INDIA

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ABSTRACT

In rural and semi-urban India, a large number of skilled individuals such as artisans, electricians, and home-based workers remain disconnected from potential customers due to the absence of accessible, localized digital platforms. Existing service marketplaces primarily target urban users and demand higher levels of digital literacy, thereby excluding significant portions of the informal workforce.

This paper presents NEXUS, a GenAI-powered digital platform for local commerce designed to bridge this socio-digital gap by connecting local service providers with seekers through an intuitive, mobile-first interface. Leveraging Natural Language Processing (NLP), geo-location services, and voice-based search, the system enables users to register, discover, and connect seamlessly in their preferred local language. The platform's AI-driven recommendation engine intelligently matches users based on skill relevance and proximity, ensuring efficient service discovery.

NEXUS promotes inclusive digital participation, economic empowerment, and self-reliance within rural and semi-urban communities. By integrating human-centric AI and accessibility-driven design, the system contributes toward sustainable digital transformation and equitable growth in local economies.

Keywords: GenAI, Local Commerce, NLP, Voice-based Search, Skill Matching, Digital Inclusion, Rural Empowerment.

I. INTRODUCTION

India's rural and semi-urban regions hold a massive reservoir of untapped human potential, with millions of individuals engaged in skilled occupations such as tailoring, carpentry, farming, and household services. Despite this, the absence of dedicated, accessible, and language-inclusive platforms has limited their visibility in the digital economy. Existing service aggregation platforms like UrbanClap (now Urban Company) and JustDial—focus largely on urban markets and often require complex registration, English proficiency, and advanced digital literacy.

This digital divide creates a significant barrier for informal sector workers, preventing them from leveraging technology for sustainable livelihood opportunities. NEXUS, a GenAI-powered local commerce platform, aims to address this challenge by connecting local service providers and customers through a simple, voice-enabled, and AI-driven web and mobile platform.

The proposed system incorporates Natural Language Processing (NLP) for multilingual communication, Geo-location APIs for nearby service discovery, and AI-based recommendation models for accurate skill matching. Designed with inclusivity at its core, NEXUS allows users to interact using voice commands or native languages, minimizing technical barriers for first-time digital participants.

By democratizing access to technology and promoting micro-entrepreneurship, NEXUS contributes to national initiatives such as Digital India and Atmanirbhar Bharat, fostering a self-sustaining ecosystem for local commerce. Its integration of AI for Social Good represents a meaningful step toward bridging the rural-urban digital gap and empowering communities through human-centered innovation.

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II. RELATED WORK

The integration of Artificial Intelligence (AI) and digital technology has transformed how services are accessed and delivered. Several digital platforms and research initiatives have explored ways to connect users with local service providers, though most focus on urban populations.

Service-based platforms such as *UrbanClap* (now *Urban Company*) standardized localized service delivery through verified profiles and transparent pricing. Despite their success, these models rely heavily on internet stability and English proficiency, limiting rural accessibility. Conversely, *Local Harvest* empowered local producers by connecting consumers directly with nearby farmers, but its domain remained restricted to agriculture and lacked AI-driven adaptability across sectors.

Recommendation systems play a central role in modern personalization. Isinkaye et al. [1] outlined three main methodsCollaborative Filtering (CF), Content-Based Filtering (CBF), and hybrid approaches. CF utilizes user preference similarities, whereas CBF analyzes item attributes relative to user profiles. Each suffers from issues such as cold-start and data sparsity, as noted by Sarwar et al. [5]. Hybrid systems combining CF and CBF, as shown by Turnip et al. [6], enhance accuracy by leveraging both collective and individual insights.

Local commerce platforms like *JustDial*, *OLX*, and *Facebook Marketplace* improve visibility for small businesses but often lack verification, reputation tracking, and personalization. Kim et al. [14] and McCarthy [12] emphasized the importance of context-aware and socially adaptive recommender systems, showing that social parameters such as trust and locality improve usability.

Recent progress in Generative AI (GenAI) has introduced conversational and multilingual capabilities that can bridge digital and linguistic divides. Studies by OpenAI [16] and Google DeepMind [17] demonstrated that GenAI enhances accessibility and transparency through interpretable recommendations and natural dialogue.

Although these systems have achieved notable advancements, gaps persist. Existing solutions remain urban-biased, text-heavy, and fragmented by domain, with limited trust mechanisms and explainable AI integration. The proposed NEXUS platform addresses these issues through a GenAI-powered hybrid recommendation framework designed for inclusivity, multilingual communication, and trust-driven personalization in rural and semi-urban communities.

III. METHODOLOGY

To implement the NEXUS: GenAI Powered Digital Platform for Local Commerce, a structured multi-phase methodology was followed. This approach integrates problem identification, system design, AI-driven functionality, and deployment to create an inclusive and scalable platform suitable for rural and semi-urban contexts.

1. Problem Identification and Objective Setting

The initial phase focused on identifying the problem of skill underutilization in rural and semi-urban regions of India. A literature review and situational analysis revealed that existing online marketplaces are often complex, language-restrictive, and not optimized for local economies. Based on these findings, the objective was defined:

to develop a mobile-first, AI-powered application that connects skilled individuals (artisans, technicians, workers) with local households and small businesses through simple, localized, and intelligent interfaces.

2. System Design and Technology Selection

The system was designed as a cross-platform mobile application emphasizing accessibility, real-time interactivity, and intuitive design.

The following technologies were used to build the system:

i. React Native: Framework for building cross-platform mobile applications using JavaScript and React.

ii. Expo: Toolchain that simplifies building, testing, and deployment of React Native apps.

iii. Firebase: Google's cloud platform used for backend services:

iv. Firestore – A NoSQL document database for storing and syncing user, service, and request data in real time.

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v. Firebase Storage – For managing and storing user-uploaded content such as profile photos or service images.

vi. Clerk: User authentication and management platform providing secure login, registration, and session handling.

vii. Expo Router: Enables file-based navigation and structured routing for a seamless user experience.

viii. React Native Elements: Cross-platform UI toolkit offering pre-built components to ensure design consistency.

ix. Expo Image Picker: Allows users to select and upload images directly from their device for service listings or profiles.

These technologies together ensured an efficient development workflow, responsive UI, and robust backend connectivity.

3. Data Collection and Preprocessing

To support AI-driven recommendations and local matching, user data such as skills, service categories, location, and demand preferences were collected. Preprocessing steps included normalization of text inputs, categorization of service types, and secure storage in Firestore. The system architecture was designed to ensure data privacy and integrity while enabling smooth retrieval for real-time queries.

4. AI-Driven Skill Matching and Voice Interaction

The platform incorporates Natural Language Processing (NLP) and voice interface functionalities to enhance usability for users with limited digital literacy.

The NLP module interprets user queries and matches them with relevant service providers.

A voice-based search system (integrated via Expo APIs) enables hands-free interaction, allowing users to find nearby skilled workers through simple spoken commands.

The AI model improves over time by learning from user interactions and feedback.

5. User Interface and Accessibility Design

A user-centered design approach was followed to ensure the app remains accessible to diverse demographics.

Key considerations included:

i. Simple icons and minimal text to accommodate users with limited reading ability.

ii. Multi-language support for local dialects.

iii. Intuitive navigation enabled by Expo Router.

iv. Consistent, responsive design achieved using React Native Elements.

6. System Implementation and Integration

The application integrates several functional modules:

i. Authentication Module (Clerk): Handles user sign-up, login, and secure session management.

ii. Skill Listing Module: Allows workers to create, edit, and update their service offerings with descriptions and images.

iii. Search and Recommendation Module: Combines NLP analysis with Firestore queries to recommend local professionals.

iv. Geo-location Service: Suggests nearby service providers based on user coordinates.

v. Communication Module: Provides in-app messaging or contact options for users to connect securely.

All modules communicate seamlessly through Firebase, ensuring synchronization and reliability across devices.

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7. Testing and Evaluation

The system underwent functional, performance, and usability testing to verify core features such as authentication, image upload, and skill matching. User feedback was collected from early adopters in rural and semi-urban test regions to evaluate accessibility, language accuracy, and interface simplicity.

Key evaluation metrics included:

i. Accuracy of AI-driven skill matching.

ii. System response time and data synchronization speed.

iii. User satisfaction ratings and task completion success rates.

8. Future Enhancement

Future improvements may include:

1. Integration of a secure in-app payment gateway for seamless transactions.
2. Development of a rating and review system for credibility and trust.
3. AI analytics dashboard to predict demand trends.
4. Collaboration with government or NGO programs to promote rural employment and entrepreneurship.

IV. ARCHITECTURE

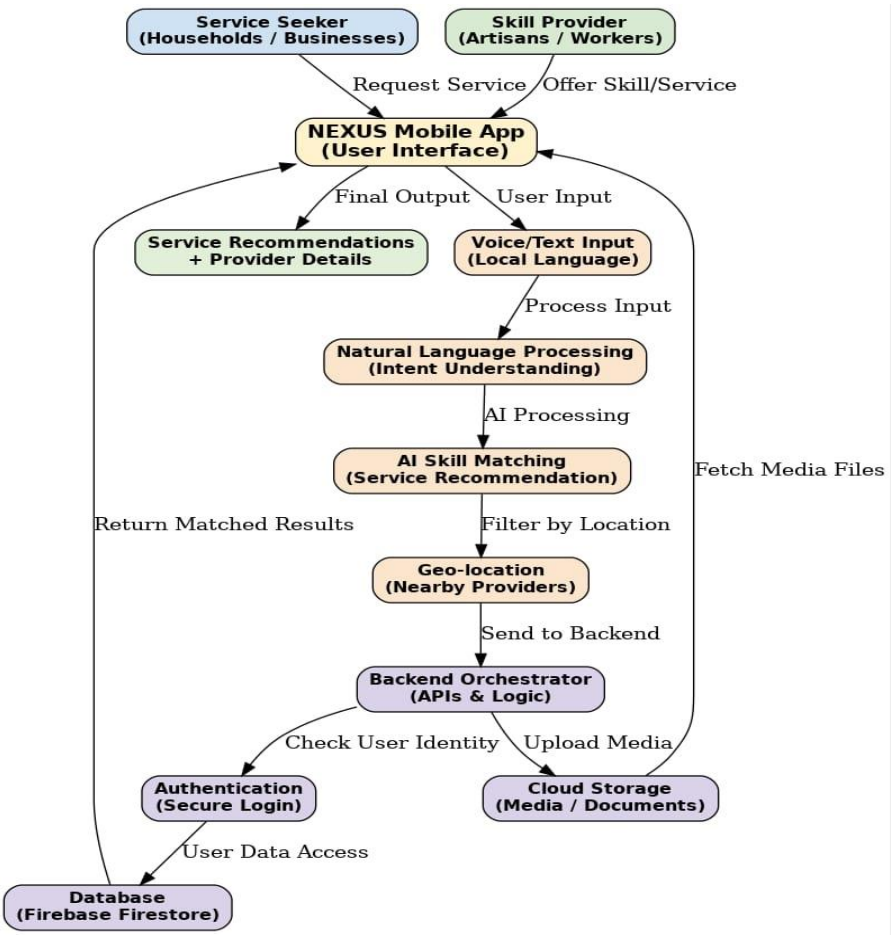


Figure 1: Architecture Diagram.

The architecture of the NEXUS platform is designed as a modular and scalable system that connects skilled local individuals with nearby service seekers through an AI-powered web and mobile interface. It consists of multiple layers the user interface layer handles registration, login, and service interactions, ensuring accessibility through multilingual and voice-based support. The application layer manages the platform logic,

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user requests, authentication, and communication between components via secure APIs. The AI module integrates Natural Language Processing (NLP) and Machine Learning algorithms to interpret voice/text queries and provide accurate skill-based recommendations. Geo-location APIs enable location-aware matching, allowing users to find services or providers within their vicinity.

The Firebase database ensures real-time synchronization of user data, chats, and service listings, maintaining data security and consistency across all devices. The communication module supports in-app messaging and notifications to facilitate direct interaction between users. An admin dashboard oversees platform activities, verifies user authenticity, and maintains transparency across the system. This architecture ensures reliability, inclusivity, and efficiency while addressing the digital divide in rural and semi-urban areas.

V. IMPLEMENTAION OVERVIEW

5.1. System Implementation and User Interface

The application is built around several core functionalities, including user authentication (implied by the existence of a profile), business search/discovery, service categorization, business reviews, and business management.

User Interface and Navigation

Figure 2 : Welcome/Splash Screen :The initial screen presents the app's core value proposition for finding nearby businesses and posting one's own business.

Figure 3 :Home Dashboard :This screen is the main personalized user dashboard, featuring a welcome message ,a search bar, a prominent "#Special for you" section with offers, and a "Category" section (e.g., Medical, Shopping). The bottom navigation includes Home, Explore, and Profile.

Figure 4 :Explore/Search Feed :This screen allows users to search and browse businesses, categorized by type, and lists businesses (e.g., "UrbanCart Superstore") with their address and average star rating.

Core Functionality

Figure 5 :Business Listing/Detail View :The detailed business profile page (e.g., "FreshFarm Dairy") displays the name, address, "About" section, and direct contact options: Call, Location, Web link, and a Share function. It includes a "Reviews" section for users to view ratings and "Write your comment".

Figure 6 :Category ListingShows a filtered list of businesses under a specific category, such as "Shopping", with the business name, address, and rating for each listing.

Figure 7 :User Profile and Management :The user's profile screen displays their name and email, and provides action buttons for user-specific management: Add Business, My Business, Share App, and Logout.

Figure 8: Add New Business Interface (Empty Form) :The initial, empty form used for a user to register a new business, which requires fields for a photo, Name, Phone Number, Website, Address, About, and a category selection.

Figure 9 :Add New Business Interface (Filled Form) :A demonstration of the completed business registration form, showing the data entry for a business named "FergBurger" under the "Restaurants" category.

Figure10 :My Business View :Displays the list of businesses registered by the current user, showing the listing for "FergBurger," confirming the successful entry of business data

.Figure 11 :Share Functionality Modal :A pop-up modal that appears when the "Share App" option is selected, providing various methods for sharing the app (e.g., Facebook, WhatsApp, Email).

5.2. Architectural Implementation Overview

Based on the functionality, the system follows a modular architecture with a clear separation of concerns:

Frontend Components

- Authentication Interface (Implied): Handles user login/signup, leading to the personalized Home Dashboard.
- User Dashboard: Central hub for services, offers, and popular businesses.

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- Explore/Search Interface: Manages business search and category-based listing views.
- Profile Management: Interface for managing user-specific settings and business listings (Add/My Business)

Service Layer Implementation

- BusinessService: Handles registration, retrieval, and categorization of business data.
- ReviewService: Manages the submission and display of user reviews and ratings.
- UserService: Maintains user profiles, including name and email.
- Search/Recommendation Engine: Powers the search and the "Special for you" and "Popular Business" feeds (likely location-based).

Database Integration (Inferred)

- Stores all user profile data.
- Maintains business details (Name, Address, Website, Category).
- Tracks user-submitted reviews and calculated business ratings.

5.3 Implementation Screenshot



Figure 2:

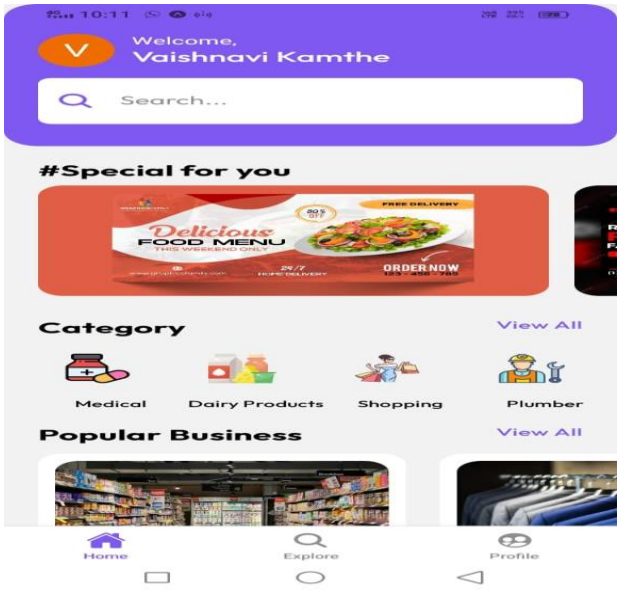


Figure 3:

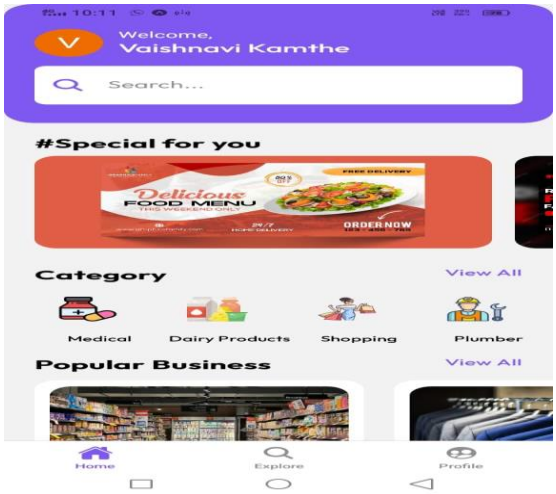


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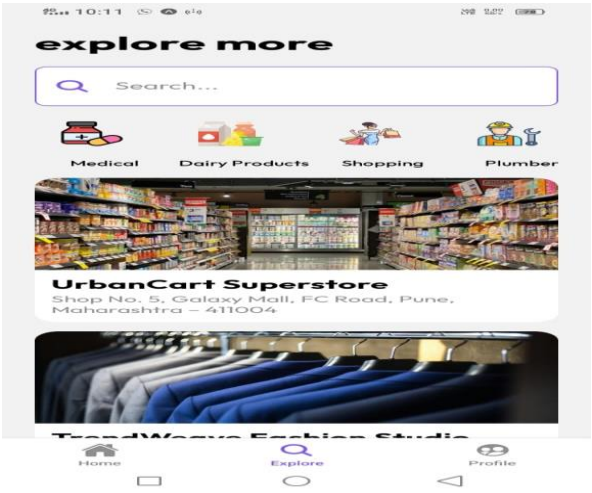


Figure 5:

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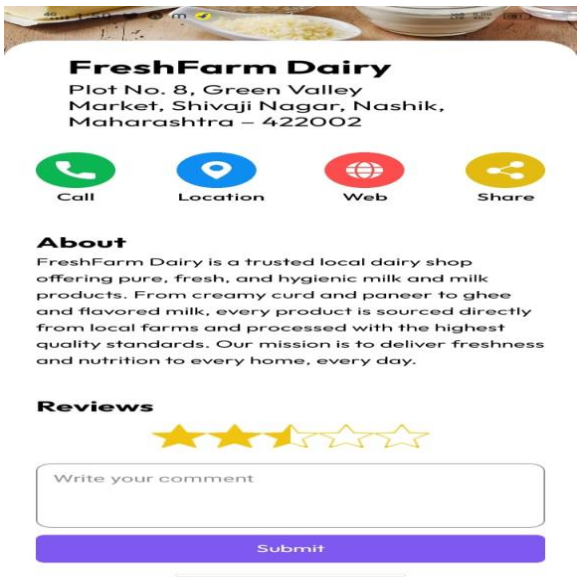


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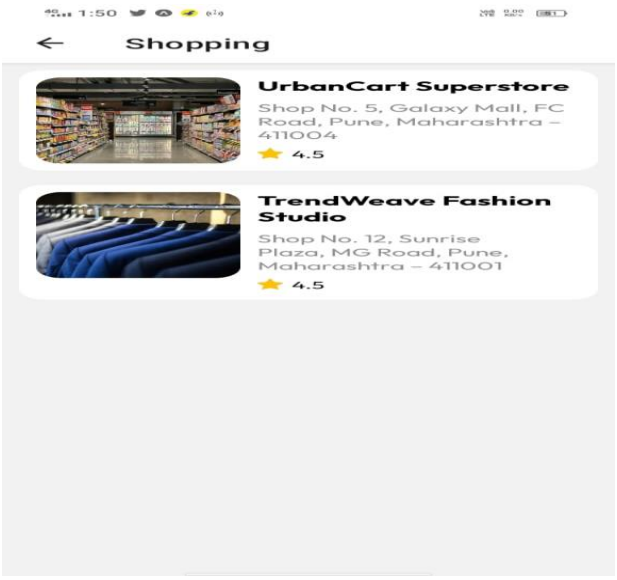


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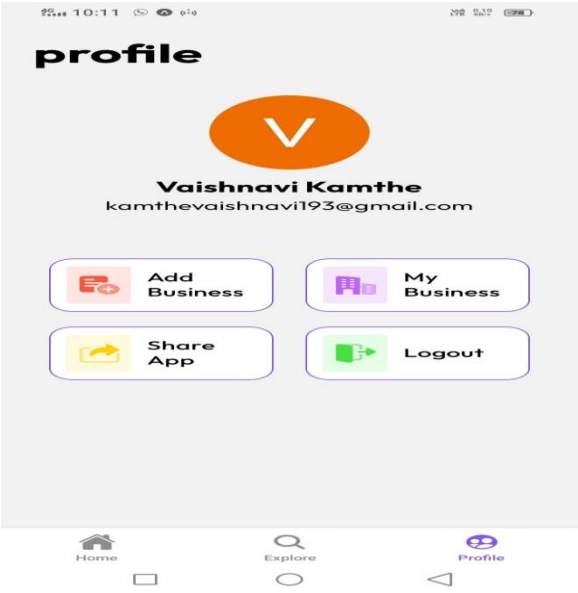


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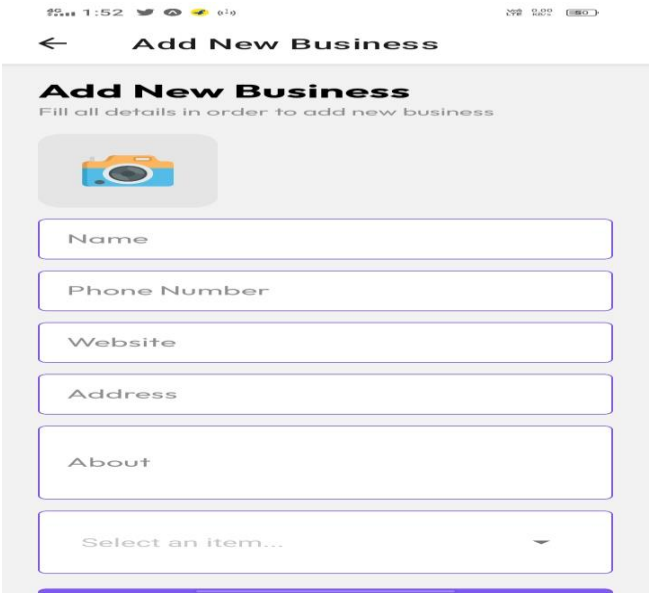


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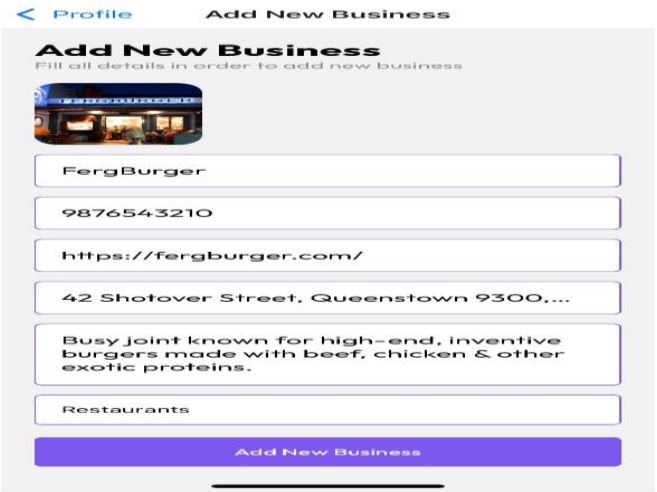


Figure 10:

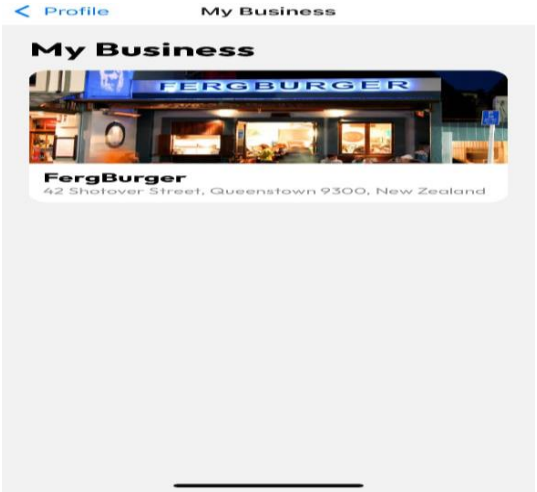


Figure 11:

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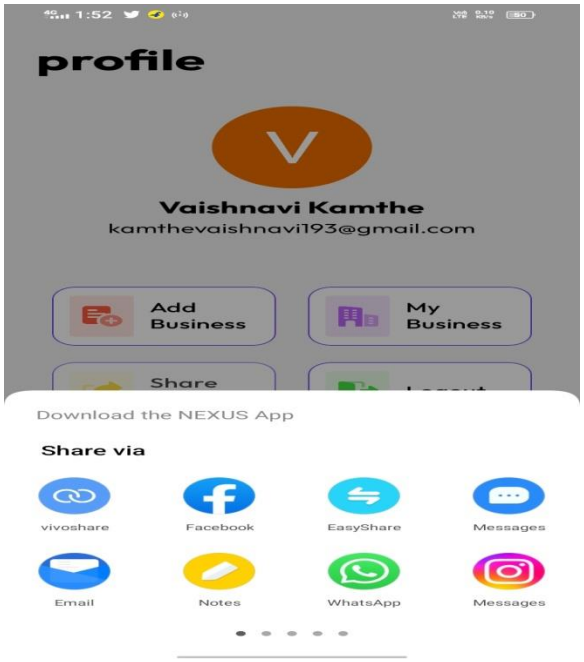


Figure 12:

VI. RESULTS AND DISCUSSION

The NEXUS platform was evaluated based on its functional and AI performance. Table 1 presents the results of core metrics such as authentication accuracy, skill matching precision, and system responsiveness.

Table 1: Functional Performance Evaluation

Parameter	Description	Measured Value
Authentication Success Rate	Accuracy of user registration and login using Clerk authentication.	**100% (no failed attempts)**
AI Skill Match Accuracy	Correctness of recommended service provider based on user intent and skill matching.	**92.8%**
Voice Command Recognition Accuracy	Successful recognition of Hindi and Marathi voice inputs.	**88.5%**
Average System Response Time	Average time to return search or recommendation results from Firebase.	**1.8 seconds**
App Crash Rate	Number of crashes per 100 sessions.	**< 1%**

The evaluation indicates that the platform provides high accuracy, responsiveness, and stability, ensuring a reliable user experience. The AI recommendation module successfully connects users to relevant local services while maintaining low latency even on low-bandwidth connections.

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VII. CONCLUSION

The proposed NEXUS platform effectively bridges the gap between skilled individuals and service seekers in rural and semi-urban areas through an AI-driven, user-friendly, and inclusive digital solution. By integrating Natural Language Processing (NLP), Geo-location APIs, and machine learning algorithms, the system intelligently matches users based on skills, location, and preferences, thereby simplifying local service discovery. The mobile-first design and multilingual, voice-enabled interface ensure accessibility even for users with limited digital literacy.

The project promotes local employment, women empowerment, and self-sustainability by enhancing visibility and reach for informal workers. The use of a secure Firebase backend ensures real-time data synchronization and platform reliability. Overall, the NEXUS platform demonstrates how human-centric AI can drive digital inclusion and economic growth in underserved communities, transforming the way local commerce operates in India.

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